Python Lab
DSF24 (Morning)
Total Marks : 100.
Time : 1.5 hours.

## Consider the following case study and answer the questions.

The Pakistan Cricket Board (PCB) aims to strengthen team selection and performance strategies
through data analysis.
As a Data Analyst in the PCB's analytics department, your task is to process player data to uncover performance patterns, identify consistent performers, and spot areas needing improvement.
You are required to validate the data, handle possible inconsistencies, and generate insights that
will help selectors make informed decisions.
Your analysis should focus on batting averages, strike rates, fitness levels, and player feedback
to enhance team efficiency and strategy planning.
You are provided with a sample dataset in the following format:

```
players_data = [
"id:1,name:Babar Azam,role:Batsman,avg:58.4,strike_rate:88,fitness:92,feedback:Excellent",
"id:2,name:Shaheen Afridi,role:Bowler,avg:12.5,strike_rate:85,fitness:88,feedback:Helpful",
….
…

]
```

Find attached .txt file.

### Q1– Player Record Conversion (Dictionary Format)

Convert each player's raw data string into a list of Python dictionaries with the following structure:

Each record now has:

- `id` (as key in the outer dict)
- `name` (string)

- `role` (string)
- `avg`, `strike_rate`, `fitness` (floats/ints)
- `feedback` (string)

Example:

```
players = [
   {
    "id": ...,
    "name": ...,
    "role": ...,
    "avg": ...,
    "strike_rate": ...,
    "fitness": ...,
    "feedback": ...
  },
  ...
]
```

**Q2 – Count Total Players:**

Count how many players are in the dataset.

**Q3 – Role-wise Fitness Summary**
Calculate the average fitness score for each role.

**Q4 – Low Fitness Identification**
Find players whose fitness is below 75%.

**Q5 – Highest Batting Average**
Find the player with the highest batting average.

**Q6 – Lowest Batting Average**
Find the player with the lowest batting average.

**Q7 – Strike Rate Above 95**
List all players whose strike rate is greater than 95.

**Q8 – Unique Roles Count**
Count how many distinct roles exist in the team.

**Q9 – Feedback Type Count**
Count how many players got each type of feedback (Excellent, Very Good, Helpful,

Average).

**Q10 – Overall Team Batting Average**
Find the overall average of all players' batting averages.

**Q11 – Fitness Grade Assignment**
Give a fitness grade:

- 90+ → A
- 80–89 → B
- 70–79 → C
- below 70 → D.

**Q12 – Strike Rate Improvement**
Assume each player's strike rate improves by 10%. Calculate the new strike rate.

**Q13 – Role Summary**
For each role, show: number of players and average fitness.

**Q14 – Strong Batsmen**
List batsmen with batting average > 50.

**Q15 – Fitness Risk Tag**
Mark players with fitness < 75 as "High Risk", otherwise "OK".

**Q16 – Sort by Batting Average**
Sort all players from highest to lowest batting average.

**Q17 – Undervalued Performers**
Players with average > 50 but feedback not "Excellent".

**Q18 – Convert Feedback to Lowercase & Count 'helpful'**
Convert all feedbacks to lowercase and count "helpful".

**Q19 – Role Balance Check**
Check if the team has at least: 2 Batsmen, 1 Bowler, 1 All-Rounder, 1 Wicket-Keeper.

**Q20 – Simple Player Summary Card**

For each player, print a small report line with name, role, avg, strike rate, fitness and feedback.